CMAN :=

Task 1 := Given that

$$S = \frac{\max(|W|)}{127}$$

$$|W| = \begin{bmatrix} 0.85, & 1.20, & 0.34, & 2.10 \\ 0.07, & 0.91, & 1.88, & 0.12 \\ 1.55, & 0.03, & 0.44, & 2.31 \\ 0.18, & 1.03, & 0.77, & 0.55 \end{bmatrix}$$

$$\max(|W|) = 2.31$$

$$\boxed{S = \frac{2.31}{127} = 0.018188}$$

Task 2 := Quantize

Given that $W_q = \text{round}(W/S)$, clamp to $[-128, \cancel{127}\ 127]$

Row 1 :=

$$\frac{0.85}{0.018188} = 46.734 \Rightarrow 47$$

$$\frac{-1.20}{0.018188} = -65.977 \rightarrow -66$$

$$\frac{0.34}{0.018188} = 18.693 \rightarrow 19$$

$$\frac{2.10}{0.018188} = 115.460 \rightarrow 115$$

Row 2 :=

$$\frac{-0.07}{0.018188} = -3.848 = -4$$

$$\frac{0.91}{0.018188} = 50.032 = 50$$

$$\frac{-1.88}{0.018188} = -103.364 = -103$$

$$\frac{0.12}{0.018188} = 6.597 \rightarrow 7$$

Row 3 :=

$$\frac{1.55}{0.018188} = 85.221 = 85$$

$$\frac{0.03}{0.018188} = 1.649 = 2$$

$$\frac{-0.44}{0.018188} = -24.191 = -24$$

$$\frac{-2.31}{0.018188} = -127.006 = -127$$

Row 4 :=

$$\frac{-0.18}{0.018188} = -9.896 = -10$$

$$\frac{1.03}{0.018188} = 56.630 = 57$$

$$\frac{0.77}{0.018188} = 42.335 = 42$$

$$\frac{0.55}{0.018188} = 30.239 = 30$$

All values fall within [-128, 127]

$$\boxed{W_q = \begin{bmatrix} 47 & -66 & 19 & 115 \\ -4 & 50 & -103 & 7 \\ 85 & 2 & -24 & -127 \\ -10 & 57 & 42 & 30 \end{bmatrix}}$$

Task 3 :=

$$W_deq = W_q \times S$$

$$S = 0.018188$$

Row 1

$47 \times 0.018188 = 0.8548$

$-66 \times 0.018188 = -1.2004$

$19 \times 0.018188 = 0.3455$

$115 \times 0.018188 = 2.0916$

Row 2

$-4 \times 0.018188 = -0.0727$

$50 \times 0.018188 = 0.9094$

$-103 \times 0.018188 = -1.8733$

$7 \times 0.018188 = 0.1273$

Row 3 :=

$85 \times 0.018188 = 1.5459$

$2 \times 0.018188 = 0.0363$

$-24 \times 0.018188 = -0.4365$

$-127 \times 0.018188 = -2.3098$

Row 4 :=

$-10 \times 0.018188 = -0.18188$

$57 \times 0.018188 = 1.0367$

$42 \times 0.018188 = 0.7638$

$30 \times 0.018188 = 0.5456$

Dequantize W_deq = W_q × S

W_deq (FP32) :=

$$W_deq = \begin{bmatrix} 0.8548 & -1.2004 & 0.3455 & 2.0916 \\ -0.0727 & 0.9094 & -1.8733 & 0.1273 \\ 1.5459 & 0.0363 & -0.4365 & -2.3098 \\ -0.18188 & 1.0367 & 0.7638 & 0.5456 \end{bmatrix}$$

Task 4 := Error Analysis :=

$|W - W_deq|$

Row 1 :=

$|0.85 - 0.8548| = 0.0048$

$|-1.20 - (-1.2004)| = 0.0004$

$|0.34 - 0.3455| = 0.0055$

$|2.10 - 2.0916| = 0.0084$

Row 2 :=

$|-0.07 - (-0.0727)| = 0.0027$

$|0.91 - 0.9094| = 0.0006$

$|-1.88 - (-1.8733)| = 0.0067$

$|0.12 - 0.1273| = 0.0073$

Row 3 :-

$|1.55 - 1.5459| = 0.0041$

$|0.03 - 0.0363| = 0.0063$

$|-0.44 - (-0.4365)| = 0.0035$

$|-2.31 - (-2.3098)| = 0.0002$

Row 4 :-

$|-0.18 - (-0.1818)| = 0.0018$

$|1.03 - (1.0367)| = 0.0067$

$|0.77 - 0.7638| = 0.0062$

$|0.55 - 0.5456| = 0.0044$

Error Matrix =

$$\begin{bmatrix} 0.0048 & 0.0004 & 0.0055 & 0.0084 \\ 0.0027 & 0.0006 & 0.0067 & 0.0073 \\ 0.0041 & 0.0063 & 0.0035 & 0.0002 \\ 0.0018 & 0.0067 & 0.0062 & 0.0044 \end{bmatrix}$$

Largest error = 0.0084 at (row1, col4)

W = 2.10, W_deq = 2.0916

Mean absolute Error (MAE) = $\frac{\text{Sum of all errors}}{16}$.

Sum of all errors = 0.0048 + 0.0004 + 0.0055 + 0.0084 + 0.0027 + 0.0006 + 0.0067 + 0.0073 + 0.0041 + 0.0063 + 0.0035 + 0.0002 + 0.0018 + 0.0067 + 0.0062 + 0.0044

Sum = 0.0696

$$\boxed{MAE = \frac{0.0696}{16} = 0.00435}$$

Task 5:= Bad Scale Experiment.

Given S_bad = 0.01 , clamp to [-128, 127]

Row 1:=

$0.85/0.01 = 85$

$\frac{-1.20}{0.01} = -120$

$\frac{0.34}{0.01} = 34$

$\frac{2.10}{0.01} = 210 \rightarrow$ clamp (127)

Row 2:=

$\frac{-0.07}{0.01} = -7$

$0.91/0.01 = 91$

$\frac{-1.88}{0.01} = -188 \rightarrow$ clamp (-128)

$\frac{0.12}{0.01} = 12$

Row 3:=

$\frac{1.55}{0.01} = 155 \rightarrow$ clamp (127)

$\frac{0.03}{0.01} = 3$

$\frac{-0.44}{0.01} = -44$

$\frac{-2.31}{0.01} = -231 \rightarrow$ clamp(-128)

Row 4:=

$\frac{-0.18}{0.01} = -18$

$\frac{1.03}{0.01} = 103$

$\frac{0.77}{0.01} = 77$

$\frac{0.55}{0.01} = 55$

W_q_bad :=

$$\begin{bmatrix} 85 & -120 & 34 & 127 \\ -7 & 91 & -128 & 12 \\ 127 & 3 & -44 & -128 \\ -18 & 103 & 77 & 55 \end{bmatrix}$$

← 127 clamped was 210

← -128 clamped was -188

← 127 clamped was (155), -128 clamped was (-231)

Dequantize : W_deq_bad = W_q_bad × 0.01

$$\begin{bmatrix} 0.85 & -1.20 & 0.34 & 1.27 \\ -0.07 & 0.91 & -1.28 & 0.12 \\ 1.27 & 0.03 & -0.44 & -1.28 \\ -0.18 & 1.03 & 0.77 & 0.55 \end{bmatrix}$$

Error = |W − W_deq_bad| :=

$$\begin{bmatrix} 0.00 & 0.00 & 0.00 & 0.83 \\ 0.00 & 0.00 & 0.60 & 0.00 \\ 0.28 & 0.00 & 0.00 & 1.03 \\ 0.00 & 0.00 & 0.00 & 0.00 \end{bmatrix}$$

Sum of errors = 0.83 + 0.60 + 0.28 + 1.03 = 2.74

$$\boxed{\text{MAE_bad} = \frac{2.74}{16} = 0.17125}$$

When S is too small, the bigger numbers don't fit in the allowed range. So they get stuck at the boundary and when we try to bring them back to original they just come out wrong